

Title: To create an 8-bit shift register using D flip flops in Logisim

Theory:

A shift register is a group of flip-flops connected in series to store and shift multiple bits of data either left or right with clock pulses.

- The registers which will shift the bits to the left are called "Shift left registers".
- The registers which will shift the bits to the right are called "Shift right registers".

The types of Shift registers which exist are

- Serial In Serial Out (SISO)
- Serial In Parallel Out (SIPO)
- Parallel In Serial Out (PISO)
- Parallel In Parallel Out (SISO)

Flip Flop is an electronic device or to be precise a kind of memory component that can hold one bit of data. A flip flop has two states, that is "SET" and "RESET". Those states are represented with the binary values 0 and 1. The flip flop remains in its current state until it receives a signal that switches it to the opposite state. A clock or pulse signal may "trigger" the flip flop to change state.

D flip flop is an electronic device that is known as "delay flip flop" or "data flip flop" which is used to store a single bit of data.

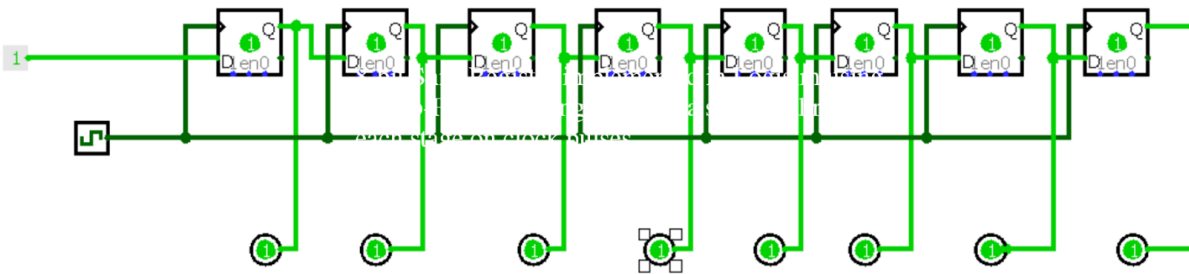
- D flip flops can be either synchronous or asynchronous.
- The clock signal is required for the synchronous version of D flip flops but not for the asynchronous one.
- The D flip flop has two inputs, data and clock input which controls the flip flop.
- When clock input is high, the data is transferred to the output of the flip flop.
- When the clock input is low, the output of the flip flop is held in its previous state.

Thus D flipflops essentially just pass on their value to their output with no processing whilst holding memory. This makes D flip flops easy to design and computationally fast allowing them to be used for the following tasks.

Registers: D flip flop is used to create a register, which can hold data in a digital system. By using the D flip flop the designer can build any size of register as per the requirement.

Counters: D flip flops are used to create the counters which count the number of events occurring in the digital system.

Implementation:



Conclusion:

Successfully implemented Shift registers , allowing it to be used as a template for future experiments which may need a register.